



HOLOTRADE

An Exchange for Compute

Node-seconds priced on live energy, node genetics,
hardware health, and fabric coherence

Every compute marketplace in existence sells the same product: a rectangle of capacity for a span of time. That product is a billing artefact — it exists because provisioning used to take minutes. When the unit boots in 171 milliseconds, the hour has no physical justification left, and everything interesting about a machine that the hour was flattening becomes priceable.

This paper is about what becomes priceable.

Wil Dahn

Built on the $W(3,3)$ - E_8 substrate programme

github.com/wilcompute/Holotrade

August 2026

How to read the evidence tags. Every headline figure in this document carries one. **PROVED** means finite mathematics with a machine-checkable proof in the repository. **MEASURED** means it came out of a run you can reproduce with `npm test`. **MODELLED** means it is a simulation parameter or an estimate, and is never presented as a measurement. **PUBLISHED** means it is somebody else's result and is cited as such. Where a figure does not survive scrutiny, the correction stays on the page beside it.

Contents

1	The thesis, in one page	4
2	What is actually being sold	6
2.1	The case against the hour	6
2.2	Three objects, not one	7
2.3	Why isolation is a precondition for the market existing at all	7
3	The price, term by term	9
3.1	E — energy	9
3.2	G — genetics	9
3.3	D — demand and wear: the balancer	10
3.4	H — health	12
3.5	Q — quantum	12
3.6	L — locality	13
3.7	The floor: what the exchange refuses to do	13
4	The network is the computer	14
4.1	The identity	14
4.2	The consequence: you buy shapes, not counts	15
4.3	The mechanism this creates: a defragmentation market	15
4.4	And the market is itself fractal	16
4.5	Joining is free	17
5	Instruments	18
5.1	Liquidity, computed rather than observed	19
6	Energy and the thermodynamic floor	20
6.1	The honest efficiency denominator	20
7	The hardware	22
7.1	It is proved, not just written	22
7.2	Thirty-nine logic cells	22
7.3	The gate refuses rather than degrades	23
8	What exists today	24

8.1	The reproducibility contract	24
8.2	Two bugs the tests caught, and why that is the point	25
9	Honest scope	26
9.1	The risks worth stating	26
A	Glossary	28
B	The integers	29

1 The thesis, in one page

Compute is the largest new commodity of the century and it is being sold with the crudest possible pricing model: a fixed hourly rate for an undifferentiated box. That model destroys information in four directions at once.

1. **It destroys the energy signal.** Wholesale electricity prices move every five minutes and go negative overnight in Texas. An hourly rate cannot express that, so the operator eats the volatility and prices in a margin to survive it. The buyer pays that margin whether or not the volatility happened.
2. **It destroys the machine's history.** Two identical servers that have run different work for a month are no longer the same product. One of them already carries statistical priors for the job you are about to run and will finish in fewer hours. Nobody prices that, because an hourly rate has no place to put it.
3. **It destroys the wear signal.** A machine everybody wants is a machine wearing out faster than its neighbours, and a machine nobody wants is still aging, still drawing standby power, and still due for the same service visit. A single rate charges both the same and balances neither.
4. **It destroys the shape.** Forty machines scattered across a datacentre and forty machines wired as one coherent block are sold as the same forty machines. They are not remotely the same thing, and the difference — bandwidth between them — is then sold back to you as a separate line item.

Holotrader prices all four. The mechanism is a single multiplicative formula in which each term is independently observable and shown to the buyer on every quote:

$$P = P_0 \times E \times G \times D \times H \times Q \times L$$

base $\times$ energy $\times$ genetics $\times$ demand/wear $\times$ health $\times$ quantum $\times$ locality

The unit that formula prices is not an hour. It is the **node-second**, made possible because a modern microVM boots in about 171 milliseconds PUBLISHED — two orders of magnitude below the settlement interval, so a one-second slice is more than 99% useful work.

The result that pays for the whole design

The demand term D is *two-sided*: it charges a premium above a target utilisation band and pays a discount below it. Run the exchange with that mechanism on and off, and the Gini coefficient of fleet utilisation — the standard measure of how unevenly load is distributed — falls from **0.164** to **0.083**. MEASURED

That is a **50% reduction in fleet dispersion**, and it is not decorative. Thermal cycling, not duty cycle, is what physically destroys silicon. A fleet whose nodes are evenly loaded cycles less, lasts longer, and produces service events spread across the calendar instead of

arriving in a clump that has to be staffed for.

Everything else in this paper follows from taking those two moves seriously: a sub-second unit, and a price with observable parts.

2 What is actually being sold

2.1 The case against the hour

Every incumbent sells node-hours. The hour is not a unit of computation; it is a unit of *scheduling overhead*. It became standard because bringing a conventional virtual machine into existence takes minutes, and if provisioning costs you three minutes then billing in anything smaller than an hour is absurd.

IN PLAIN ENGLISH: What a virtual machine is

A physical server is one piece of hardware. A *virtual machine* is a software-created illusion of a whole separate computer running on top of it, so that one physical server can be rented to several customers at once without them being able to see or interfere with each other. The software that creates the illusion is called a *hypervisor*. Creating a traditional virtual machine involves booting a complete operating system inside the illusion, which is why it takes minutes.

IN PLAIN ENGLISH: What a microVM is, and why it changes the economics

A *micro VM* is a stripped-down virtual machine: a real hypervisor boundary, a real separate operating-system kernel, but with everything unnecessary for one specific job removed — no virtual graphics card, no virtual sound card, no legacy device emulation. It gets the same security isolation as a full virtual machine in a small fraction of the startup time. Amazon's *Firecracker* is the best-known implementation and is what runs their serverless products.

Published measurements put microVM boot at a median of **171.5 ms**, with 95% under 176 ms and 99% under 178 ms, measured on Firecracker over rotational storage — a deliberately pessimistic baseline, since production fleets use flash **PUBLISHED**. Once the unit comes into existence in under a fifth of a second, three things follow immediately.

1. **The hour has no physical justification.** A one-second billing slice carries roughly 17% startup overhead in the worst case where every second is a fresh boot, and effectively zero for any job that runs longer than a few seconds. There is no technical reason to round up to 3,600 seconds.
2. **Per-second energy pricing becomes possible rather than theatrical.** You cannot price a resource at a granularity finer than your settlement interval. If you settle hourly, the five-minute grid price is information you have thrown away before you could act on it.
3. **The startup cost becomes visible instead of smeared.** Holotrader charges the boot *once, explicitly, as its own line*, rather than folding it into the rate. A customer running many short jobs can therefore see exactly what their fan-out pattern costs them — which is information the hourly model structurally cannot provide.

2.2 Three objects, not one

Layer	Object	What it is
ASSET	the node	A durable machine with an identity, an accumulated learning history, and a physical wear record. This is what you own, lease, or lend into the pool.
CONTRACT	the execution plan	A cryptographically signed document naming exactly what will run, against which software artefacts, with which network permissions, inside which time window. It exists — and can be traded — before anything executes.
UNIT	the node-second	What actually settles. Metered against the real electricity price at the second it was drawn.

IN PLAIN ENGLISH: What an execution plan is, and why it matters commercially

Think of it as a signed work order. Before any machine starts, the plan states: here is the exact software (identified by a cryptographic fingerprint, so it cannot be swapped for something else), here is the only network address it is permitted to contact, here is the window during which this authorisation is valid, and here is my signature over all of it.

Commercially this is the difference between renting a machine and buying a *guarantee about what happened on it*. It is what lets a bank, a hospital, or a defence contractor use somebody else's hardware. It is also what makes compute genuinely tradeable: a signed plan is a contract with a specification, so it can change hands before it runs, the way a futures contract does.

2.3 Why isolation is a precondition for the market existing at all

A two-sided marketplace requires that an owner of idle machines be willing to run a stranger's code on them. Without hardware-enforced isolation, “list your spare capacity” is a request to be compromised, and the supply side never materialises.

The isolation model Holotrader assumes is the strong one now standard in this class of system: each job gets its own operating-system kernel under a real hypervisor (not a shared kernel, as with containers); the guest has *no network device at all* by default, with all input and output crossing a single narrow channel to the host; the filesystem is cryptographically sealed so that tampering prevents boot rather than being detected afterwards; and every lifecycle event is written to a hash-chained log PUBLISHED.

IN PLAIN ENGLISH: What a hash chain is

Each entry in the log includes a fingerprint of the entry before it. Change or delete any entry and every subsequent fingerprint stops matching — so tampering is not merely discouraged, it is arithmetically detectable by anyone holding the log. It is the same idea that underpins

a blockchain, without needing a blockchain.

3 The price, term by term

The formula is multiplicative rather than additive. That is a deliberate engineering choice with three consequences: the terms stay independent so each can be measured on its own; each keeps its own auditable line on the quote; and because each is separately bounded, no single factor can drag the price to zero or to infinity by itself.



each term independently observable, each shown on the quote

3.1 E — energy

IN PLAIN ENGLISH

Datacentres buy electricity on a wholesale market where the price changes every few minutes. In Texas it goes *negative* at three in the morning — the grid pays you to consume — and spikes to four figures during a heat wave. E is the term that passes that reality through to the buyer.

E is deliberately *not* linear in the grid price. A node should not become worthless when the grid goes negative, nor unsellable when it spikes. Holotrader applies a power law with exponent 0.55 on the ratio to that site’s baseline, clamped to [0.62, 2.40]. That reproduces how operators actually behave: they pass through most of a normal move and none of a tail. The adjustment also accounts for PUE .

IN PLAIN ENGLISH: What PUE is

Power Usage Effectiveness. For every watt that reaches a computer chip, a datacentre spends additional watts on cooling, power conversion and lighting. PUE is the ratio of total facility power to computing power. A PUE of 1.0 would be perfect; a well-run modern facility is around 1.1, and a hot-climate facility with mechanical chillers is 1.4 or worse. It is a straight multiplier on the electricity bill and therefore on the price.

3.2 G — genetics

This is the term that makes Holotrader a different product rather than a better price list.

IN PLAIN ENGLISH

When a machine runs the same class of work repeatedly, the software layer on it accumulates useful statistical structure — cached compilations, tuned kernel selections, warm model checkpoints, learned scheduling parameters. The industry already exploits this deliberately: “warm starts” and “checkpoint merging” are standard practice in machine-learning operations.

The consequence is that two physically identical servers, after a month of different work, are *not the same product*. One of them will reach your target in fewer hours. Nobody prices

that today. Holotrade does, and calls the accumulated structure the node's *genome* because it is inherited when a machine's software state is copied onto fresh hardware.

G is built from three separately observable quantities, never from a claim:

- **Specialisation** on *this specific* workload class — not a general capability score. A node with excellent memory-bandwidth characteristics gets no credit for it on a workload that is compute-bound.
- **Realised fitness**: the completion record. Jobs finished versus jobs failed, weighted by experience depth. Never a manufacturer's specification.
- **Provenance depth**: a long lineage of successful forks is evidence that a software core is genuinely good rather than lucky.

The caveat that goes on every lease quote

A leased core keeps specialising toward whatever *you* feed it. Over a 90-day lease it will measurably drift toward your workload class and away from the one it was strongest at when you rented it. You are buying a starting point, not a frozen asset — and by the end of a long lease you will have changed the thing you rented. Holotrade prints the projected drift on the quote rather than leaving the customer to discover it.

3.3 D — demand and wear: the balancer

IN PLAIN ENGLISH

The obvious way to price demand is one-sided: a busy machine costs more. That is what surge pricing does, and it leaves the cold half of your fleet earning nothing while still aging, still drawing standby power, and still due for the same maintenance visit.

Holotrade charges a premium above a target band *and pays a discount below it*. The discount is not generosity. It is cheaper than the alternative, and the reason is physical.

Why an evenly loaded fleet physically lasts longer

Semiconductor devices fail predominantly from *thermal cycling*, not from accumulated running time. Every time a chip heats up and cools down, the materials in it expand and contract at different rates; solder joints, die attachments and interconnects accumulate mechanical fatigue with each cycle. This is standard reliability engineering and is why the failure curve is modelled with a Weibull distribution whose shape parameter is above 1, meaning wear-out rather than random failure.

So a machine held steadily at 70% load wears *more slowly* than one swinging between idle and pinned at the same 70% average. Reducing the *dispersion* of load across a fleet therefore does three things at once: it uses the capital better, it extends the physical life of the hardware, and it spreads service events across the calendar instead of clustering them.

IN PLAIN ENGLISH: What a Gini coefficient is

A single number between 0 and 1 measuring how unevenly something is distributed. Economists use it for income inequality. Here it measures how unevenly work is distributed across the fleet: 0 would mean every machine carries exactly the same load, 1 would mean one machine does everything while the rest sit idle. Lower is better.

The measured effect, and why it is a finding rather than an animation

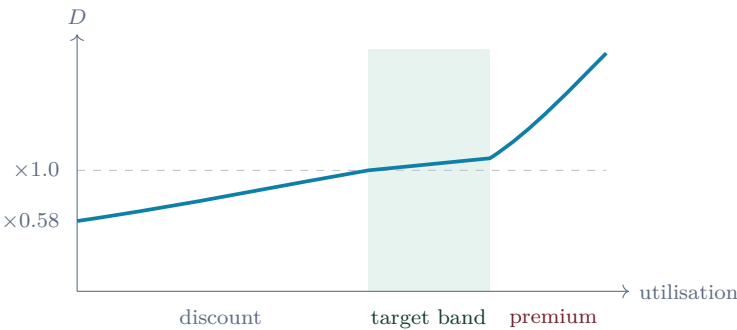
	Balancer off	Balancer on
Utilisation Gini	0.164	0.083

Nothing in the simulation pushes utilisation toward the band directly. A node’s demand target is set by its *price* relative to the median of its own hardware class, and that price was itself computed from its utilisation. The band is where the feedback loop settles, not where it is aimed. Turn the balancer off in the running application and the dispersion climbs. **MEASURED**

Two engineering details matter for whether this is honest.

The comparison set is the hardware class, not the fleet. A buyer choosing between accelerators does not consider a general-purpose CPU node “cheap” — they are not substitutes. Comparing against a fleet-wide median would make every CPU look like a permanent bargain and every scarce node look permanently extortionate, and the loop would never settle.

The curve must be monotone. If there is any utilisation at which a node gets *cheaper* by becoming busier, a buyer can arbitrage the discount by loading the machine they are about to rent. An earlier version of this design had exactly that defect at one of the seams between the discount, band and premium regions. It was caught by a monotonicity test, and the test is in the repository.



The premium region is superlinear because the wear term is: a node that swings is accumulating damage faster than a node that is merely busy, and the premium it earns is the maintenance reserve being funded by the people causing the wear. That is not arbitrage. It is the correct allocation of a real cost.

3.4 *H* — health

A worn machine delivers less per hour, so it should cost less per hour, and it should be *visibly* cheaper rather than quietly slower. *H* multiplies three observable things: the performance derate you actually receive, a reliability discount derived from the instantaneous failure hazard, and a drift term from accumulated correctable memory errors.

IN PLAIN ENGLISH: What a correctable error is, and why it predicts failure

Server memory detects and silently repairs single-bit corruptions. A machine reporting a rising count of these is not yet failing, but it is a well established leading indicator that it will. Holotrade treats the count as a priced signal rather than an alert nobody reads.

3.5 *Q* — quantum

IN PLAIN ENGLISH

This term is 1 — exactly, permanently — for every ordinary workload. It is included because the underlying architecture makes a sharp distinction between two kinds of computation, and that distinction turns out to be commercially useful even for customers who will never buy a quantum anything.

There is a class of operations, called *Clifford* operations, that a theorem of Gottesman and Knill shows can be simulated on any ordinary computer in polynomial — that is, practical — time **PUBLISHED**. On this architecture, that class covers everything structural: the routing, the memory addressing, the error correction, the fault tolerance. All of it runs on any hardware anyone already owns.

What is genuinely scarce is the other kind: *non-Clifford* operations, sometimes called “magic”. Each one multiplies the cost of classical simulation by a factor of 9. At $t = 6$ that is 531,441; at $t = 20$ it is 1.2×10^{19} , and no classical fleet on Earth can help you.

Why this matters even if you never buy quantum capacity

It gives the product a knob nobody else has. The *structure* of the system is free and universal — every machine can run it. The *power* is a continuously priced resource. You provision advantage the way you provision storage: as much as you pay for, on a substrate everyone already runs.

And the refusal is hard. A workload requiring non-Clifford operations cannot be served by ordinary hardware *at any price* — not slowly, not approximately. Holotrade refuses the order and states the reason rather than silently substituting something that will not work. That refusal is enforced in software, tested, and also implemented in the hardware primitive described in §7.

3.6 L — locality

L prices how far a candidate machine is from where your data already sits. Every provider gestures at this with “availability zones”, which are administrative labels rather than measurements. Here it is an exact distance, and §4 explains why it can be computed in a handful of gates with no routing table at all.

3.7 The floor: what the exchange refuses to do

A two-sided pricing mechanism with an aggressive discount needs a floor, or the balancer will happily discount a machine to a price that never repays it.

$$\text{floor} = \text{electricity} + \text{maintenance reserve} + \text{capital recovery}$$

All three components, and the third is the one usually omitted. Electricity for a CPU node is a few cents an hour; a floor of energy alone would permit a discount that never repays the machine. *Capital recovery* is the purchase price of the chassis spread across the hours it will actually deliver, derated by health — because a worn machine has fewer useful hours left over the same calendar and must recover its capital faster.

IN PLAIN ENGLISH: What a maintenance reserve is

Money withheld from today’s revenue to pay for a service visit that has not happened yet. Rising steeply with the machine’s failure hazard, which means a popular machine — which is wearing faster — is putting more aside. That is the honest way to charge the premium: the people causing the wear fund the repair.

The exchange **refuses to clear below the floor**, and there is a test asserting it across the entire fleet on every quote. When a quote is sitting at the floor, the interface says so explicitly: “*the balancer wanted to discount this node further and the exchange refused.*”

4 The network is the computer

This section explains the one genuinely unusual idea in the product, and why it produces a market mechanism that has no analogue elsewhere.

4.1 The identity

Holotrade’s machines are addressed by their position in a mathematical structure called $W(3, 3)$: the symplectic generalized quadrangle over the four-dimensional space on three symbols. It has 40 points and 240 connections.

IN PLAIN ENGLISH: What that structure is, without the jargon

Forty positions, each labelled by four digits where every digit is 0, 1 or 2. Two positions are directly connected if and only if a specific short arithmetic formula on their labels comes out to zero. That is the entire definition. The resulting network has some remarkable properties that were established by mathematicians early in the twentieth century for reasons having nothing to do with computing.

The properties that matter commercially, all of which are **PROVED** and machine-checked in the repository:

Property	Value	Why a buyer cares
Diameter	2	Any machine can reach any other in at most two hops, <i>regardless of the traffic pattern</i> . Hyperscale datacentre fabrics need three to six hops under load.
Alternative paths	4	Any two non-adjacent machines share exactly four independent two-hop routes. Four-way redundancy with zero configuration.
Bisection	100 of 240	Cut the network in half any way you like and 42% of all connections cross the cut. This is the number that governs how fast a large job can move data internally.
Fault tolerance	11 / 5	Survives 11 machines crashing, or 5 machines actively lying, without losing agreement.
Routing table	none	The forwarding decision is one arithmetic test on the address. No table to build, distribute, converge, or keep fresh.

“Address is route”

In a conventional network, deciding where to send something requires consulting a routing table — a data structure that has to be constructed, distributed to every device, agreed upon, and continuously updated as things change. The protocols that do this (BGP, OSPF) are a large fraction of the complexity, the failure modes and the operating cost of a modern

network.

Here the decision is:

$$\langle u, v \rangle = u_0v_1 - u_1v_0 + u_2v_3 - u_3v_2 \pmod 3$$

and two machines are adjacent exactly when that is zero. The address *is* the route. §7 shows this compiled to 39 logic cells.

4.2 The consequence: you buy shapes, not counts

Because routing and computing are the same operation on this structure, **compute and bandwidth are one provisioned quantity**. And that means the thing being sold is a *shape*, not a count.

	40 scattered machines	40 machines as one cell
What it is	40 computers, plus a network bill to make them cooperate	One computer
Worst-case hops	whatever the network gives you	2
Bisection	near zero	exactly 100
Redundant paths	configure them	4 , free

Same machine count. Completely different product. So Holotrade prices *coherence* — how much of the achievable internal bandwidth a basket actually holds — and prices it at the **basket** level, never the node level, because coherence is not a property any individual machine has.

The relationship is superlinear: the last few connections that complete a cell are worth more than the first few, because they are what collapse the diameter and unlock the redundant paths.

The whole really is worth more than the sum of its parts

And here that is a measurable quantity rather than a slogan. It is the realised bisection of what you actually hold, divided by the bisection you would have had if you had spent the same money on contiguous capacity. **PROVED**

4.3 The mechanism this creates: a defragmentation market

IN PLAIN ENGLISH

Trading breaks shapes. After a few months of ordinary spot buying, every participant holds confetti — a few machines in each of forty different blocks. Each individual position looks perfectly fine. Collectively, the market’s total usable bandwidth has collapsed, and nobody can see it happening from inside their own position.

The fix is a **swap book**. I give you my isolated machine sitting in *your* block; you give me yours sitting in *mine*. No cash changes hands beyond a small adjustment for the difference in condition between the two machines. Both of us end up with more coherent holdings and the market’s total delivered bandwidth rises.

It is disk defragmentation, except that the value being recovered is bandwidth — and unlike a disk, both parties can be made strictly better off, which is what makes the swaps voluntary.

A swap is only offered when both sides gain

This is enforced, not aspirational. The engine computes the counterparty’s own coherence gain before proposing anything, and there is a test asserting it. There is no version of this where the venue improves one book at another’s expense — because in that version nobody takes the other side, and the mechanism dies.

4.4 And the market is itself fractal

IN PLAIN ENGLISH

A network of computers, wired this particular way, behaves as a single computer. So a network of *those* is also a single computer. And so on upward, indefinitely. This is not a metaphor about scale — it is a substitution rule: replace each of the 40 positions with an entire copy of the whole structure, and everything that was true of the original is true of the result.

The scaling is exact: a level-*n* network has 40^{*n*} machines at a worst-case distance of 8*n* hops.

PROVED

Level	Machines	Worst-case hops	Seats
1	40	8	a rack
2	1,600	16	a building
3	64,000	24	a campus
4	2,560,000	32	a city pilot
5	102,400,000	40	a nation
6	4,096,000,000	48	half the planet
7	163,840,000,000	56	everyone, and every device

The commercial consequence: one order book at every scale

Because a composed network satisfies exactly the same interface as a single machine — it has an address, an accumulated history, a health record, a utilisation, a throughput — **the same pricing engine quotes both, and cannot tell the difference.**
An operator lists their entire campus as a single instrument. A buyer sees it as one line in

the same order book where an individual accelerator appears. Orders at one level aggregate into liquidity at the level above. **The book is self-similar because the machine is,** and there is a test in the repository that quotes a leaf, a cell and a campus through one code path and asserts all three decompose identically.

This also changes what a service-level agreement is. “Worst case 16 hops” is not a promise someone makes and pays penalties for missing. It is a *theorem about the shape you bought*.

4.5 Joining is free

A machine joins by being grafted into the structure at one position. Nothing else in the network has to be told, no rebalancing pass runs, no distributed agreement is needed. Because the structure’s symmetry group acts transitively on positions, the new machine is immediately indistinguishable from every other.

For a two-sided marketplace this is the difference between “list your idle hardware” costing nothing and costing a fleet-wide event — which is exactly where distributed-hash-table and blockchain-based compute markets have historically bogged down.

5 Instruments

Five, because the risks a compute buyer actually carries are five different risks. Only the first is a machine rental.

Instrument	Tenor	The risk it addresses
Spot node-hour	immediate	You need capacity now and will accept today's price.
Forward block	1–90 days	You need capacity in March and cannot carry the risk of a Texas heat dome tripling your electricity input. Priced as spot plus cost-of-carry, with a risk premium scaled by that specific site's historical volatility.
Burst option	1–30 days	You <i>might</i> need 400 machines on launch day. You want the right, not the obligation. Priced off utilisation volatility — the probability the capacity is actually free times the size of the spike you are insuring against.
Genome lease	7–365 days	You want a machine whose software core <i>already knows your workload class</i> . Priced on realised fitness and specialisation rather than on silicon, with a term discount because a long commitment de-risks the operator's utilisation.
Supply offer	open	You own idle machines. The exchange prices them, routes work to them, and pays you the clearing price less the maintenance reserve.

Where the durable margin is

Spot node-hours are a commodity, and commodities converge to marginal cost — which is precisely the floor the exchange refuses to clear below. **There is no durable business in selling undifferentiated node-hours**, and any business plan that assumes otherwise is planning to lose a price war.

Trained cores are not a commodity. They are heterogeneous, their quality is observable and verifiable, and their value *compounds with use*. A machine that has run genomics workloads for six months is a strictly better genomics machine than the identical chassis next to it, and that gap widens. That is where the spread lives, and it is why the genome lease is a first-class instrument rather than a footnote on the spot product.

5.1 Liquidity, computed rather than observed

IN PLAIN ENGLISH

Normally you find out how liquid an asset is by watching it trade. That is backward-looking and useless for a new listing. Here it can be computed in advance from the constraints attached to the asset.

Every real-world constraint is a symmetry the asset *does not have*. A machine pinned to one jurisdiction by data-residency law cannot be moved by the transformations that would otherwise move it. A bare-metal reservation is fixed harder still. An air-gapped enclave hardest of all.

Each constraint shrinks the set of positions at which a counterparty could legitimately take delivery — and that set *is* the addressable market for that specific asset. Formally it is an orbit under the structure’s symmetry group, and the orbit size is the group order divided by the size of the subgroup that holds the asset fixed.

Constraint	Orbit	Co-volume	Band
none	40	1.000	deep
GPU affinity	27	0.667	liquid
single-tenant	20	0.500	liquid
data residency	13	0.333	thin
air-gapped	5	0.125	bilateral

Why this is worth having

It is the honest, checkable reason a residency-pinned machine should not trade at the same price as an identical unpinned one — and it is a *number the buyer can verify* rather than a discount the seller asserts. For a venue, it also produces an inventory metric: a book whose holdings are mostly “bilateral” has an inventory problem, not a marketing problem, and the dashboard says so.

6 Energy and the thermodynamic floor

IN PLAIN ENGLISH

Compute is bought in dollars and paid for in joules. Data centres drew about 415 terawatt-hours globally in 2024 — roughly 1.5% of world electricity — growing at 12% a year, more than four times the growth rate of total electricity demand. The International Energy Agency projects this more than doubling to about 945 TWh by 2030, comparable to Japan's entire national consumption PUBLISHED.

That projection assumes incremental improvement to current architectures. It is therefore a statement about a technology trajectory, not a physical law — and an exchange that prices electricity per second is the mechanism that makes the distinction visible to a buyer, because it is the first place where an efficient machine can actually be rewarded for being efficient at the moment it is efficient.

6.1 The honest efficiency denominator

IN PLAIN ENGLISH: The Landauer limit

Physics sets a hard floor on how little energy it can cost to erase one bit of information: about $kT \ln 2$ joules, where k is Boltzmann's constant and T is temperature. Not an engineering target — a thermodynamic law, derived by Rolf Landauer in 1961. Operations that do *not* erase information have no such floor at all, a result due to Charles Bennett.

On this architecture almost everything is reversible and therefore erases nothing. The single irreversible step is error-correction syndrome extraction: 58 measurements per cycle, each costing $kT \ln 3$. At room temperature that is

2.64×10^{-19} joules per cycle at 300 K

PROVED

A current-generation accelerator dissipates on the order of 10^{-15} joules per operation — roughly **four to seven orders of magnitude** above that floor, depending on device class.

Why this is the right number to publish

Every efficiency claim in the industry uses a denominator someone chose — last year's product, a competitor's product, a benchmark with known biases. This denominator is a *law*. It cannot be gamed, it does not change with the product cycle, and it is the same number for every vendor.

Holotrader publishes each machine's distance from it, in orders of magnitude, on the quote. It is the most honest efficiency figure a datacentre can state.

Honest scope. This is a thermodynamic *lower bound*, not a measured device efficiency. Real hardware dissipates far more through optical loss, detector inefficiency and classical control overhead. The gap is a statement about fundamental limits and architectural headroom. It is not a claim that

any machine achieves it, and no full build of the architecture that would approach it exists.

7 The hardware

The exchange’s hot path is small enough to put on a chip, and doing so converts the central architectural claim from an assertion into an artefact.

On a venue that reprices its entire fleet every second, two operations dominate: computing the fabric distance for the locality term, and deciding whether a given plan may run on a given machine. Both are implemented in `rtl/holotrade_admit.v`.

7.1 It is proved, not just written

IN PLAIN ENGLISH: What formal verification is

Testing checks a program on the examples you thought of. *Formal verification* proves a property holds for *every possible input*, using a solver that searches exhaustively for a counterexample and reports that none exists. For a circuit with a manageable number of inputs this is not approximate — it is a proof.

Two independent implementations of the routing arithmetic were written deliberately differently — one as staged modular operators, one as plain integer arithmetic with a single reduction at the end — and proved equivalent across the **entire 65,536-input space** by a SAT solver:

SAT proof finished - no model found: SUCCESS!

Proving the two agree is necessary and *not sufficient* — both could agree and both be wrong about the geometry they claim to implement. So the software test suite closes the loop from the other side, modelling the circuit’s exact bit encoding and checking it against the actual $W(3,3)$ structure over all 1,600 ordered pairs of positions.

7.2 Thirty-nine logic cells

Synthesised to an iCE40 HX8K field-programmable gate array:

39 × SB_LUT4 **6** × SB_CARRY

MEASURED

That is the complete routing and admission decision — including the five-condition security gate and the migration cost calculation — in thirty-nine logic cells, with no table to build, distribute, converge or keep fresh.

What is deliberately not reported

No clock frequency. A cell count is a fact about a netlist; a frequency is a fact about a *specific part after place-and-route*, and this design has been neither placed nor routed. Quoting an $F_{\max}$ here would be a modelled number dressed as a measurement, and this document does not do that.

7.3 The gate refuses rather than degrades

Five conditions, checked in priority order, each reporting a distinct code rather than a bare zero:

#	Refusal	Meaning
1	BAD_SIG	The signature does not match the contents. Outranks everything — a tampered plan is never reported as merely out-of-window.
2	REPLAY	This authorisation has been used before.
3	WINDOW	Outside the validity period.
4	PIN_DRIFT	A software artefact no longer matches the fingerprint it was signed against. This closes a real supply-chain hole: a mutable tag that moved between signing and boot.
5	NO_MAGIC	The workload needs capability this machine does not have, at any price.

There is no best-effort path through this logic. A silent fallback is how a security posture becomes decorative, and putting the refusal in hardware makes it non-negotiable.

8 What exists today

Component	Status	Evidence
Substrate geometry & routing	PROVED	45 tests; SAT proof; exhaustive cross-check
Six-term pricing engine	MEASURED	runs live; floor never breached, tested fleet-wide
Two-sided balancer	MEASURED	Gini 0.083 vs 0.164, reproducible from seed
Coherence & swap book	MEASURED	both-sides-gain enforced and tested
Recursive composition	PROVED	one engine quotes leaf, cell and campus
Signed plans & admission gate	MEASURED	five refusal codes, hash-chained audit log
Per-second metering	MEASURED	settles on node-seconds, reports joules drawn
Five instruments	MEASURED	all price off the same spot quote
Trading terminal	MEASURED	12 views, no build step, runs from a static file
Hardware primitive	PROVED	39 LUT4 + 6 carry, formally verified
Physical photonic substrate	—	does not exist ; not required by anything above
Real market data feeds	—	seed data; engines are feed-agnostic by construction

8.1 The reproducibility contract

Every claim in this document that carries a **PROVED** or **MEASURED** tag can be checked by cloning the repository and running two commands:

```
npm test
npm run verify:rtl
```

45 tests: substrate facts and engine contracts
SAT-prove the routing primitive

The simulation uses a seeded pseudo-random generator, so every run reproduces exactly. There is a test asserting that too.

Two kinds of test, deliberately not mixed

Substrate facts are exact finite mathematics with single correct answers. Any deviation is a bug, not a tuning question.

Engine contracts are invariants that must hold whatever the parameters are: the floor is never breached, a capability requirement is never silently substituted, the audit chain detects tampering, the recursion closes.

Nothing in the suite asserts a tuned constant. A test that fails when a coefficient is retuned is a test that stops anyone improving the model, so the assertions are bounds and invariants,

not fixtures.

8.2 Two bugs the tests caught, and why that is the point

Both were found by the suite during development and both are the kind that survive casual review.

A doubled bandwidth figure. The bisection calculation divided by the wrong edge count, reporting exactly twice the correct value for every basket. It was caught by a test asserting that a complete cell hits precisely the known theoretical bound of 100. A plausible-looking wrong number had been passing through the pricing engine.

A non-monotone price curve. The demand multiplier had a discontinuity at one of the seams between its discount, band and premium regions, creating a narrow range of utilisation where a machine got *cheaper by becoming busier*. That is directly arbitrageable: load the machine you are about to rent, then rent it at a discount. A monotonicity test caught it, and the fix is documented in the source beside the code it corrects.

9 Honest scope

This section exists because a document that only argues one direction is not worth the paper. Every claim here is sorted by what actually supports it.

Exact and computed in this repository

The $W(3,3)$ geometry — 40 points, 240 edges, 40 lines of 4 points each, diameter 2, bisection exactly 100 as the spectral bound met by an explicit cut. The symmetry group order 51,840. The fractal law 40^n machines at $8n$ hops. The Landauer floor 2.64×10^{-19} J/cycle at 300 K. The 9^t scaling of classical simulation cost. The venue capacity identities. The routing primitive's formal equivalence proof.

Every one is checkable by running the test suite. None depends on any physical theory being correct.

Modelled, and never presented as measured

The fleet, the wholesale grid prices, the accumulated genomes, the wear trajectories and the order flow are all simulation. This is a working model with a seeded generator, not a production system with customers.

The architecture is deliberately feed-agnostic: replace the seed data file with a real grid-price API, real datacentre telemetry and a real hardware inventory, and *no engine above it changes*. That is a design property, but it is not the same as having done it.

Asserted by the source programme, not by this product. The $W(3,3)$ - E_8 substrate programme makes claims about $W(3,3)$ being a candidate physical substrate for fundamental physics. **Holotrader does not depend on any of that being true.**

Everything the exchange prices — the geometry, the routing rule, the error correction rate, the thermodynamic floor — is finite mathematics that holds whether or not the physics programme is correct. The structure is a good network topology and a good addressing scheme on its own merits, established independently of any physical interpretation.

The photonic hardware that would realise the quantum layer does not exist. What runs here is the classical simulation of the structural layer, which is polynomial-time and portable to any computer — and which is exactly why the structural layer costs nothing and the quantum advantage stays a separately priced dial rather than a capability being claimed.

9.1 The risks worth stating

- **Supply-side cold start.** A two-sided market needs machines before it needs buyers. The isolation model makes lending safe and the splice operation makes joining free, but neither creates the first thousand machines.
- **The genome premium must be demonstrated, not argued.** The claim that a specialised core finishes in measurably fewer hours is plausible, grounded in existing practice, and *not yet measured on real workloads*. It is the single most important thing to validate, because it is where the durable margin lives. Until it is measured on production jobs, it is a hypothesis

with a good mechanism.

- **Per-second settlement is an operational commitment.** Metering every running instance every second against a live grid price is real infrastructure with real failure modes, and the reconciliation problem when a meter disagrees with an invoice is not solved by having the right formula.
- **Regulatory surface.** Forward contracts and options on a physical commodity attract commodity-derivatives regulation in most jurisdictions. The spot and lease products likely do not; the forward and option products likely do, and jurisdiction by jurisdiction.

A Glossary

Term	Meaning
Adjacency	Two machines being directly connected. Here, determined by one arithmetic test on their addresses rather than by a lookup table.
Bisection	Cut a network into two equal halves; the bisection is the number of connections crossing the cut. It governs how fast a large job can move data internally, and is the number that dominates fabric cost.
Clifford operations	A class of operations provably simulable on ordinary computers in practical time (Gottesman–Knill). Covers everything structural in this architecture, and is therefore free.
Coherence	How much of the achievable internal bandwidth a collection of machines actually holds. A property of a <i>basket</i> , never of an individual machine.
Content-addressed	Identified by a cryptographic fingerprint of the contents rather than by a name. Renaming is impossible and substitution is detectable.
Correctable error	A memory corruption the hardware silently repaired. A rising count is a well-established leading indicator of impending failure.
Derate	The gap between a machine’s nameplate performance and what it actually delivers today, after wear.
Diameter	The worst-case number of hops between any two machines in a network.
Firecracker	Amazon’s open-source microVM hypervisor, which runs their serverless products.
Gini coefficient	A number from 0 to 1 measuring how unevenly something is distributed. Used here for load across a fleet; lower is better.
Hash chain	A log in which each entry commits to a fingerprint of the previous one, so removing or altering any entry breaks every entry after it.
Hypervisor	Software that creates and isolates virtual machines on physical hardware.
Landauer limit	The thermodynamic minimum energy to erase one bit, about $kT \ln 2$. A physical law, not an engineering target.
MicroVM	A stripped-down virtual machine with full hypervisor isolation but only the devices one job needs. Boots in milliseconds rather than minutes.
Node-second	Holotrader’s atomic unit: one machine, for one second, metered against the electricity price at that second.
Non-Clifford / magic	The operations that are genuinely hard to simulate classically. Each multiplies simulation cost by 9. The only scarce resource in this architecture.
Orbit	The set of positions an asset can legitimately occupy under the structure’s symmetries. Its size is that asset’s addressable market.
Pin drift	A software artefact no longer matching the fingerprint it was signed against — a real supply-chain attack surface, closed by re-verifying at admission rather than only at fetch.

Term	Meaning
PUE	Power Usage Effectiveness: total facility power divided by computing power. A straight multiplier on the electricity bill.
Splice	Grafting a new machine into the fabric. A purely local operation requiring no global reconfiguration.
Symplectic form	The short arithmetic expression on two addresses whose vanishing defines adjacency. The whole routing decision, in one formula.
Thermal cycling	Repeated heating and cooling. The dominant physical wear mechanism in semiconductors — more damaging than sustained running time.
Weibull hazard	The standard statistical model for wear-out failure. A shape parameter above 1 means failure rate rises with accumulated wear.

B The integers

Every quantity in the architecture is generated by nine integers, which are in turn forced by a single equation, $q! = 2q$, whose only positive solution is $q = 3$.

Value	Symbol	Role in the exchange
3	q	the radix; three states per digit
2	λ	common neighbours of an adjacent pair
4	μ	redundant paths between any non-adjacent pair
12	k	connections per machine
40	v	machines per cell; positions in the address shell
240	$ E $	connections per cell
100	—	bisection: the bandwidth a complete cell delivers
51,840	$ \text{Aut} $	symmetry group order; canonical address cells
27/80	—	error-correction rate cap on logical throughput
1/10	—	the contextual fraction; the attestation signal
30	$h(E_8)$	settlement latency floor, in milliseconds
58	$n - k$	irreversible measurements per cycle; the energy floor